Typed Judgment, Deterministic Enforcement: A Consent-Gated Gateway Architecture for Inter-Agent Collaboration over Messaging Channels

Yizhang Cai, Jun Liu, Di Ma, Ke Yu, Ke Li

School of Artificial Intelligence, Beijing University of Posts and Telecommunications
Beijing Guoshu Zhilian Technology Co., Ltd.
2025111259@bupt.cn

The boundary we study

Messaging events are unstructured and may be duplicated, delayed, or replayed; agent protocols expect authenticated endpoints, structured requests, and task state.

- Free-text delegation lets model output become an invocation.
- Static rules cannot resolve natural-language ambiguity.

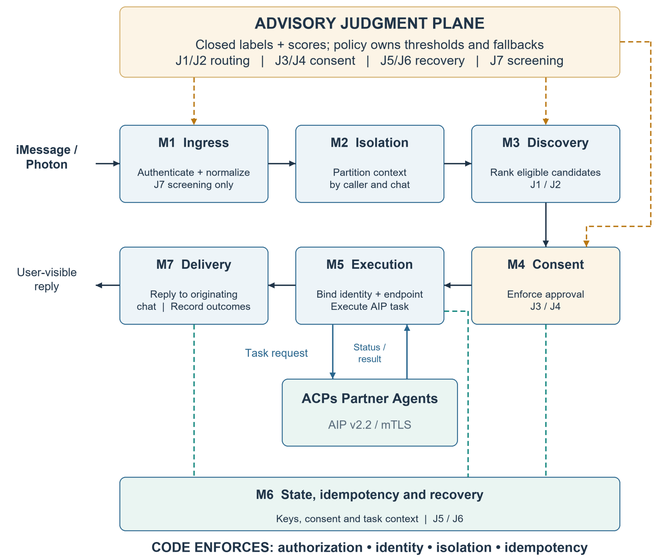
Design rule. The model answers closed questions; policy code owns thresholds and effects.

Seven modules, four invariants

M1 ingress authentication
M2 caller / conversation isolation
M3 discovery and routing
M4 consent and policy
M5 identity-bound execution
M6 state, idempotency, recovery
M7 delivery and telemetry

In code: authorization • identity • isolation • idempotency

Architecture



M1 ingress, M2 partition by caller and conversation, M3 discovery, M4 consent, M5 identity-bound execution, M6 state, M7 delivery.

Results: routing (130 requests, 3 deliveries)

Direct	48/48/50 of 50
Ambiguous	39/38/38 of 40
Out of scope	0 of 40

Results: invariants (paired ablations)

Arm	Control	Removed
Identity binding	0/30	30/30
Caller isolation	0/30	30/30
Idempotency	30 exec.	60 exec.

Same-code controls; each arm changes exactly one check.

Takeaway

Model advises; code controls identity, state, and every side effect.

Consent **0**/**40** before, **36**/**36** after. Judgment **94.1%** vs **83.9%**; Brier **0.095** vs **0.266**.

Artifact: `github.com/ankelas/consent-gated-gateway-artifact`

S&T Program of Xiongan New Area (No. XA2026111001K).

LaTeX TikZposter